

EXPERIENCE

• **Monadical**

Quebec, CA, Remote

Senior Full Stack Engineer

2022 - 2026

- **Payments Backend and Financial Ledger:** Principal author of the payments backend of a music royalty platform distributing revenue across artist networks (Node.js, Express, Neo4j, Prisma). Built a custom double-entry accounting ledger on PostgreSQL, graph-based revenue routing with event sourcing and CQRS, and Stripe and PayPal integration. Wrote the architecture decision records covering the beneficiary graph, payment-event reliability, the accounting design and routing-graph versioning
- **Event Ingestion Pipeline:** Built the path carrying Stripe, PayPal and Firebase events into the payments platform: a Node.js producer publishing persistent messages to a RabbitMQ exchange on a confirm channel with Stripe signature verification, and a consumer with per-source schema parsers, exhaustive dispatch on a tagged source, a semaphore serialising handlers, dead-letter and retry queues, reconnect-on-close and Sentry reporting at fatal level
- **Durable Execution and LLM Workflows:** Drove significant parts of a Celery to Hatchet durable-execution migration on an open-source AI meeting transcription platform (Python backend, Next.js front end): made workflow execution mandatory for multitrack processing, added DAG task models and live status broadcasting, and wrote a CLI that renders workflow runs as a text DAG. Extracted a central S3 storage abstraction on aioboto3 with S3-compatible endpoint support (Garage) for standalone deploys, integrated local LLM inference via Ollama with structured output, and shipped full-stack transcript search
- **Crawler Concurrency and Go Contribution:** Owned the browser pool, proxy routing and egress subsystem of a Playwright headless-Chromium crawler (Python, asyncio, Pub/Sub, Prometheus): backpressure in the Pub/Sub consumer through a bounded queue and nack signalling with labelled metrics, browser contexts refcounted and keyed by a hash of proxy credentials, and a second acquisition tier renting remote browsers over CDP. Second-highest contributor to a Go service computing emissions for AI model inference
- **Infrastructure and Observability:** Top committer on the payments platform's infrastructure repository: sole author of every Kubernetes and Helm path (per-app deployments, services and ingresses, a Grafana and Loki logging stack), Terraform per-app modules against DigitalOcean and Cloudflare, Ansible host inventories, and GitHub Actions driving apply and deploy. Principal contributor to the product monorepo's CI/CD: per-PR preview deploys, affected-project test runs, Sentry release tagging
- **Correctness and Testing:** Typed functional TypeScript across production code (Effect, fp-ts, io-ts), property-based testing with fast-check inside the payments codebase covering payout isomorphism and allocation, circular-dependency and type-check gates in CI, Playwright end-to-end suites
- **Client-Facing Delivery:** Worked directly with the client's team for three years on requirements, turning operational problems into technical proposals over calls and written documents inside a PRD then ADR decision process; coordinated two junior developers and onboarded two successive senior developers as the team changed

• **DesignerAdvantage**

Newton, MA, US, Remote

Senior Frontend Engineer (contract)

2021 - 2022

- **React Platform Stabilization:** Stabilized a React codebase: clearer state management and data flow, react-query for server state, fewer recurring bugs. Consumed Salesforce API data in the user interface. Set up end-to-end tests on pull requests and Sentry monitoring

• **FiddleQuest**

Rogue River, OR, US, Remote

Founding Engineer (contract, intermittent)

2014 - 2024

- **Music Education Platform, First Commit to Wind-Down:** Wrote the first commit and remained the top contributor across a decade of production for a violin teaching platform (Meteor, React, TypeScript). Owned Stripe subscription billing with dunning sequences, plus a serverless reconciliation job (AWS Lambda) that diff's the entire Stripe customer set against application state and flags orphaned customers and missing subscriptions
- **Media Pipeline, Auth and Event-Driven Services:** Built direct-to-S3 uploads with pre-signed URLs and a video transcoding pipeline (AWS Elastic Transcoder) producing watermarked output and generated thumbnails over CloudFront. Implemented an OAuth2 authorization server so a separate game service and a forum authenticate against the main app, and decomposed gameplay events across SNS, SQS and DynamoDB. Ran the infrastructure (Ansible, systemd, nginx, MongoDB, Redis) and carried out framework migrations on a live paying customer base

OPEN SOURCE & SELECTED PROJECTS

- **[huly-mcp](#)** (TypeScript, Effect, 2026) : MCP server and CLI giving AI coding agents a typed contract over the Huly platform. Published on npm and the official MCP registry with contributions merged from six outside developers. Ships a discovery-based tool surface (category listing, search, schema fetch, invoke) so a very large tool surface stays usable by clients that eagerly list tools, and a behavioral oracle so runtime-library upgrades fail CI before release.

- **Hapsland** (TypeScript, Effect, foldkit, Bend2, 2026) : review layer for coding agents in the Codex CLI edit loop. Changed declarations in self-contained TypeScript files are judged by an external classifier model against configurable type-design rules, and findings go back to the agent as advice before it finishes. Review-flow policy written in Bend2 and checked against the TypeScript implementation by differential replay of generated traces; a foldkit dashboard renders a simplified review flow directly from the compiled Bend2 transitions and computes no decisions of its own.
- **dalph** (TypeScript, Effect, 2026, in development) : graph-native delivery orchestrator for coding agents. Derives the runnable frontier from a task DAG and supervises bounded concurrent workflows with typed worktree, review, retry and recovery semantics. Protocol specified in Quint with temporal model checking, bound to the implementation by model-based conformance tests.

SKILLS

- **Languages:** TypeScript (expert), JavaScript, Python, Go (contributor), Rust, PHP (Zend Certified), Java (Oracle Certified), Scala, Elm, Bend2
- **Back-End & Data:** Node.js, Express, Python (asyncio) services, REST and typed OpenAPI clients; PostgreSQL and Prisma, schema design and migrations, MongoDB, Neo4j, DynamoDB, Redis; RabbitMQ, Pub/Sub, SQS and SNS, event sourcing, CQRS, durable execution (Hatchet)
- **Infrastructure & Observability:** Docker, Kubernetes and Helm, Terraform against DigitalOcean and Cloudflare, Ansible, GitHub Actions; Sentry, Prometheus metrics, Grafana and Loki
- **Cloud:** AWS (S3, Lambda, SNS, SQS, DynamoDB), GCP and Firebase, DigitalOcean
- **AI Engineering:** Claude Code, Codex and opencode used daily; MCP server authoring, tool-surface design for LLM consumption, agent orchestration, evaluation harnesses with deterministic replay, visualization techniques for validating agentic output, local LLM integration with structured output
- **Testing & Correctness:** property-based testing (fast-check), model-based testing against Quint specifications, Effect-native testing with typed error channels and injected layers, Playwright, vitest and jest; typed functional TypeScript (Effect, fp-ts, io-ts)
- **Front-End:** React (10+ yrs), Next.js, Vite, Tailwind; client state across observable, atomic and statechart models (MobX, jotai, XState); server state with React Query and RTK Query

EDUCATION & CERTIFICATIONS

- **Togliatti State University** Togliatti, Russia
Bachelor in Applied Informatics (Business Informatics) 2014 - 2018
- **Certifications:** [System Analysis](#) and [Asynchronous Architecture](#) (tough-dev.school); Oracle Certified Professional, Java SE; Zend Certified Engineer, PHP

PUBLICATIONS & TALKS

- [How to stay in control of your vibed codebase](#); [Emerging Rust GUI Libraries in a WASM World](#) (Hacker News front page); [Property-Based Testing for Temporal Graph Storage](#); [TypeScript Validators Jamboree](#); talks at the Montréal TypeScript and JavaScript meetup on Effect.ts, property-based testing with fast-check, and exhaustive matching with ts-pattern; internal talks on Quint, system architecture, and immutability

Keyword from the Zerion job description: WATERMELON